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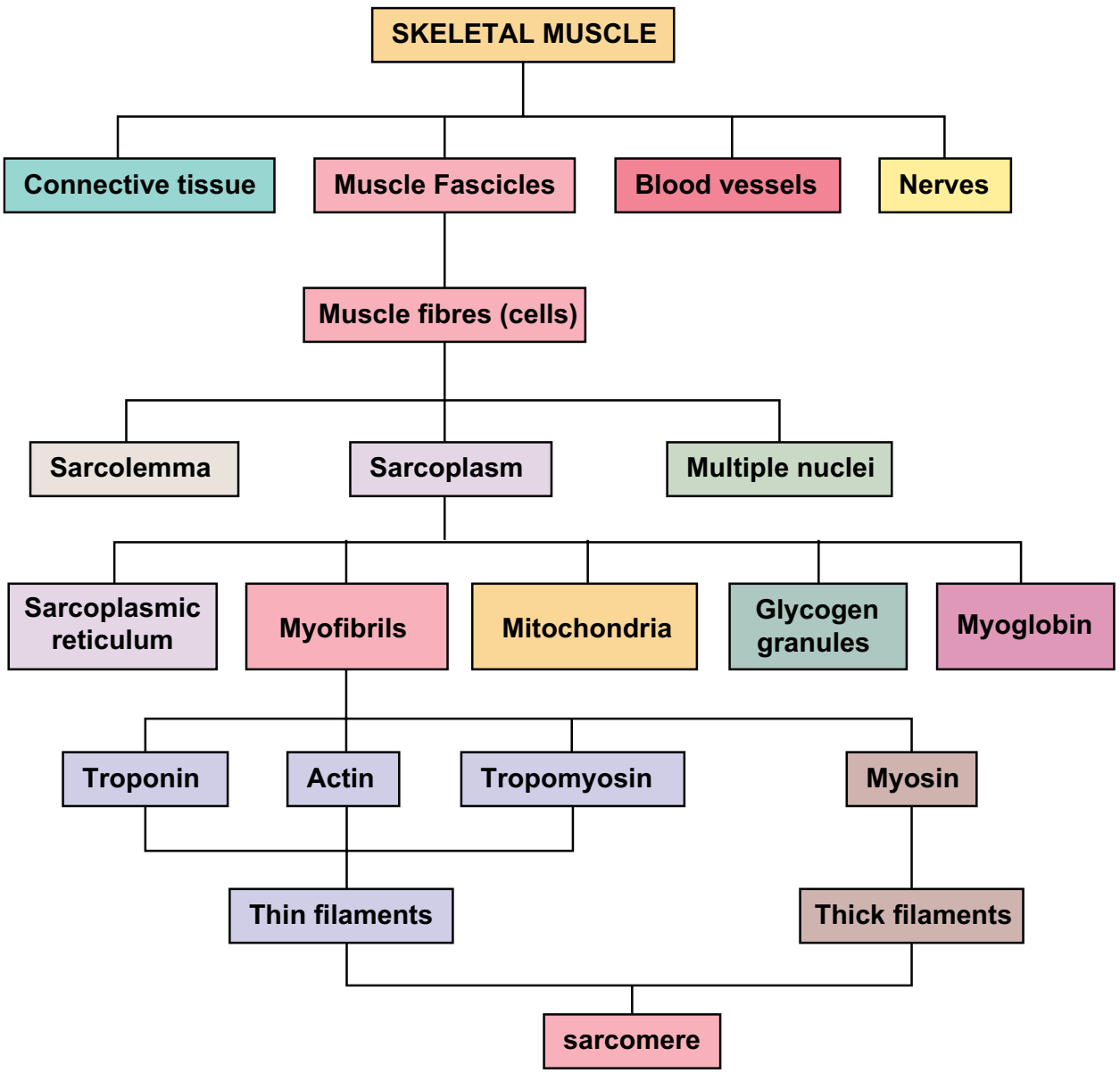
myosin molecule is made up of a monomer called meromyosin. The meromyosin has two regions, a globular head with a short arm and a tail. The short arm constitutes the heavy meromyosin (HMM). The tail portion forms the light meromyosin (LMM). The head bears an actin-binding site and an ATP- binding site. It also contains ATPase enzyme that split ATP to generate energy for the contraction of muscle. The thin filaments are composed of two interwined actin molecules. Actin has polypeptide subunits called globular actin or G–actin and filamentous form or F-actin. Each thin filament is made of two F-actins helically wound to each other. Each F-actin is a polymer of monomeric G-actins. It also contains a binding site for myosin. The thin filaments also contain several regulatory proteins like **tropomyosin** and **troponin** which help in regulating the contraction of muscles along with **actin** and **myosin**.

The study of muscle is called **myology.**

Schematic representation of organizational levels of skeletal muscle.